A Probabilistic Model of Motor Subtype Progression

Weeks 5–7 Progress Report

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Abstract

Weeks 5–7 called for a baseline probabilistic model of subtype transition, validated on held-out patients. Three were built and compared: the discrete Markov chain the proposal originally sketched, a continuous-time chain that respects the actual gap between visits, and a hidden Markov model in which a latent phenotype is observed through a misclassification matrix.

The two ordinary chains are both misspecified, and a direct test says why. In the observed labels, the next state depends strongly on the *previous* one even after conditioning on the current one — for patients currently labelled TD, arriving from PIGD raises the chance of being PIGD next visit from 0.086 to 0.393 ($\chi^2 = 156.6$, $p = 7.9 \times 10^{-33}$). A Markov chain in the label cannot represent that. A latent state read through noise can, and does: the hidden Markov model beats the discrete baseline on held-out patients in **10 of 10** independent splits, by $+0.028 \pm 0.009$ log-likelihood per transition (paired t-test $p = 5.4 \times 10^{-6}$).

The fitted model delivers the output table the proposal specifies — the probability of occupying each subtype at 1, 2, 3 and 5 years — and makes two substantive claims: the underlying phenotype is far more stable than its label (TD persists about 10 years latent against 3.4 observed), and progression is *sequential*, passing through Indeterminate rather than jumping from TD to PIGD.

1 Objective

The proposal assigns Weeks 5–7:

- Develop a simple probabilistic model (Markov or transition-probability) from the observed longitudinal transitions.
- Estimate, for each current subtype, the probability of remaining stable or transitioning at the next follow-up.
- Validate the model using held-out patients.

with the deliverable stated as an *initial progression model providing probabilities of subtype transition.* All three are complete. The model that survives validation is not the one originally sketched, and Section 3 explains why.

2 Design

2.1 Data

Input is `outputs/visit_panel.csv` from Weeks 1–2: 4,441 labelled visits over 1,080 sporadic PD patients, yielding 3,361 consecutive transition pairs from the 934 patients with at least two labelled visits.

2.2 Held-out split

Splitting is at the level of the *patient*, never the visit. A patient contributes several transitions; splitting those independently would put the same person on both sides and make the held-out estimate optimistic.

Split	TD	Indet.	PIGD	Patients	Transitions
Train	543	76	136	755	2,359
Test	233	33	59	325	1,002

Table 1: Patient-level split (70/30, seed 42), stratified by baseline subtype: 71.9% TD on the training side against 71.7% on the test side. The manifest is written to `outputs/model/split_manifest.csv`.

Of these, 658 training and 276 test patients have the two or more labelled visits a sequence model requires.

3 Why a Markov chain in the label is not enough

A Markov chain assumes that the current state is a sufficient summary of the past: $P(\text{next} \mid \text{current}, \text{previous}) = P(\text{next} \mid \text{current})$. That is testable directly, on the 2,427 labelled triples in the panel.

Currently	Triples	χ^2 (dof 4)	p
TD	1,583	156.6	7.9×10^{-33}
Indeterminate	235	15.3	4.2×10^{-3}
PIGD	609	61.2	1.7×10^{-12}

Table 2: Test of the Markov property. Rejected in every state.

Concretely, for the 1,583 visits currently labelled TD:

Arrived from	→ TD	→ Indet.	→ PIGD
TD	0.849	0.065	**0.086**
Indeterminate	0.528	0.176	0.296
PIGD	0.482	0.125	**0.393**

Table 3: Where a patient came from changes the probability of moving to PIGD by a factor of 4.6, even though they are in the same state now.

This is the signature of a latent state read through noise. If a patient's true phenotype is PIGD but one examination scores them TD, the next examination is likely to score PIGD again — producing exactly this dependence on history. It is also consistent with the Weeks 3–4 finding that 35.2% of observed subtype changes reverse at the very next visit.
The consequence is that a chain fitted to the observed label is misspecified *whatever* its time parameterisation. That prediction is tested in Section 5 and confirmed.

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4 The three models

4.1 Discrete Markov chain — the sketched baseline

One transition matrix, applied per visit, interval ignored. Six free parameters. This is the model the proposal describes as "the probability of remaining stable or transitioning at the next follow-

up".

4.2 Continuous-time Markov chain

A generator matrix Q whose off-diagonal entries are instantaneous rates. The probability of moving from state i to j over elapsed time t is

$$P(t) = \exp(Qt),$$

so every observed pair contributes at its own actual interval and the model can be evaluated at any horizon. Fitted by maximising $\sum_{\text{pairs}} \log P(\Delta t)[i, j]$ over six log-rates. Weeks 3–4 found transition probabilities scale strongly with the inter-visit gap, so this was the expected improvement.

4.3 Hidden Markov model — latent state plus misclassification

A latent phenotype $X(t)$ evolves as a continuous-time chain with generator Q. What is recorded at a visit is not X but a noisy reading Y, related by a misclassification matrix

$$E[a, b] = P(\text{label recorded is } b \mid \text{true state is } a).$$

The likelihood of a patient's whole sequence marginalises over every latent path, computed by the forward algorithm. Fourteen free parameters: six rates, six emission terms, two for the initial distribution.

This is the discrete-state counterpart of the proposal's stated observation model, *observed ratio = true ratio + measurement noise*.

An identifiability constraint. Left unconstrained the maximum-likelihood solution is degenerate, and identically so across all eight random restarts: the latent Indeterminate state emits uniformly (0.333/0.333/0.333, i.e. its label carries no information) and PIGD becomes absorbing at the parameter boundary. Requiring each state to be recorded correctly more often than not ($E_{aa} \geq 0.5$) removes both. It costs nothing: the held-out advantage over the discrete baseline is +0.027 with the constraint against +0.026 without. A sweep over floors from 0.33 to 0.80 changes that advantage only between +0.023 and +0.027.

5 Held-out results

Model	Log-lik / transition	Accuracy	Brier
HMM (latent + misclassification)	−0.6384	**0.769**	**0.3458**
Discrete Markov (per-visit)	−0.6653	0.768	0.3625
Continuous-time Markov	−0.6752	0.766	0.3670
Marginal (null)	−0.8633	0.625	0.5185

Table 4: 276 held-out patients, 1,002 transitions. Higher log-likelihood and lower Brier are better.

5.1 The continuous-time chain does not help

This is a negative result and worth stating plainly: the Weeks 3–4 finding that transition probabilities scale with the interval was correct, but correcting for it inside a chain fitted to the observed label does not improve prediction. The reason is Section 3 — the constraint $P(t) = \exp(Qt)$ forces consistency across intervals that a noisily observed process does not have, so imposing it on a misspecified model hurts rather than helps.

Interval	n	CTMC	DTMC	Difference
≤ 1.5 yr	890	-0.6534	-0.6488	-0.0046
1.5–2.5	71	-0.8151	-0.7907	-0.0244
> 2.5	41	-0.9054	-0.8061	-0.0993

Table 5: Held-out log-likelihood per transition by interval. The continuous-time chain loses at every interval and by most at long ones — where respecting the gap should have helped most.

Model	Mean	SD
HMM	$\mathbf{-0.6781}$	0.0251
Discrete Markov	-0.7064	0.0250
Continuous-time Markov	-0.7113	0.0257
Marginal (null)	-0.8764	0.0187

Table 6: Held-out log-likelihood per transition over 10 independent patient-level splits, each refitting every model from scratch.

5.2 Robustness

The HMM beats the discrete chain in **10 of 10** splits, by $+0.0284 \pm 0.0094$ log-likelihood per transition. Paired t-test $p = 5.4 \times 10^{-6}$; Wilcoxon signed-rank $p = 0.002$. The ranking HMM $>$ discrete $>$ continuous-time $>$ null does not change in any split.

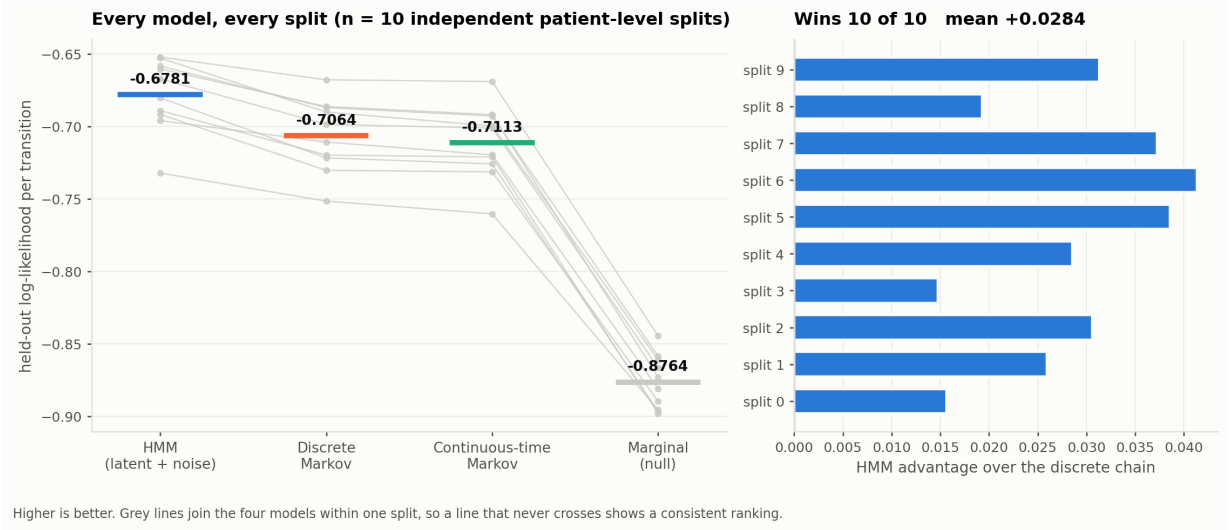


Figure 1: Every model on every split. Grey lines join the four models within one split, so a line that never crosses shows a ranking that is consistent rather than an artefact of averaging.

6 What the fitted model says

6.1 Latent rates

From	→ TD	→ Indeterminate	→ PIGD
TD	−0.0994	0.0937	**0.0057**
Indeterminate	0.1156	−0.4713	0.3557
PIGD	**0.0000**	0.0605	−0.0605

Table 7: Latent generator Q, instantaneous rates per year.

Progression is sequential. The direct TD → PIGD rate is 0.0057 per year against 0.0937 for TD → Indeterminate — **16 times smaller** — and the direct PIGD → TD rate is estimated at exactly zero. The model was not told the states are ordered; it recovered that from the data.
This is what one should expect if the underlying quantity is the continuous TD/PIGD ratio: to move from above 1.15 to below 0.90 a patient must pass through the intermediate band. The same chain fitted to the observed label puts the direct TD → PIGD rate at 0.0583, ten times higher, because apparent direct jumps are misclassification.

6.2 How reliable is the clinical label?

True state	recorded TD	recorded Indet.	recorded PIGD
TD	**0.941**	0.038	0.021
Indeterminate	0.250	**0.500**	0.250
PIGD	0.106	0.052	**0.842**

Table 8: Estimated misclassification matrix E. The Indeterminate row sits at the imposed floor of 0.50, so it should be read as "at most half of Indeterminate labels are correct" rather than as a point estimate — the unconstrained likelihood drives it to 1/3.

This corroborates a Weeks 3–4 result obtained by a completely different route: changes *into* Indeterminate were the ones that reversed most often (56.0% for TD → Indeterminate, 50.0% for PIGD → Indeterminate). Two independent analyses agree that the Indeterminate label is the least reliable of the three.

6.3 The phenotype is more stable than its label

State	Observed label (CTMC)	Latent phenotype (HMM)
TD	3.41	**10.06**
Indeterminate	0.44	**2.12**
PIGD	2.28	**16.52**

Table 9: Mean years before leaving a state. Treating the label as the state makes every phenotype look transient; separating measurement error from genuine change recovers a much slower underlying process.

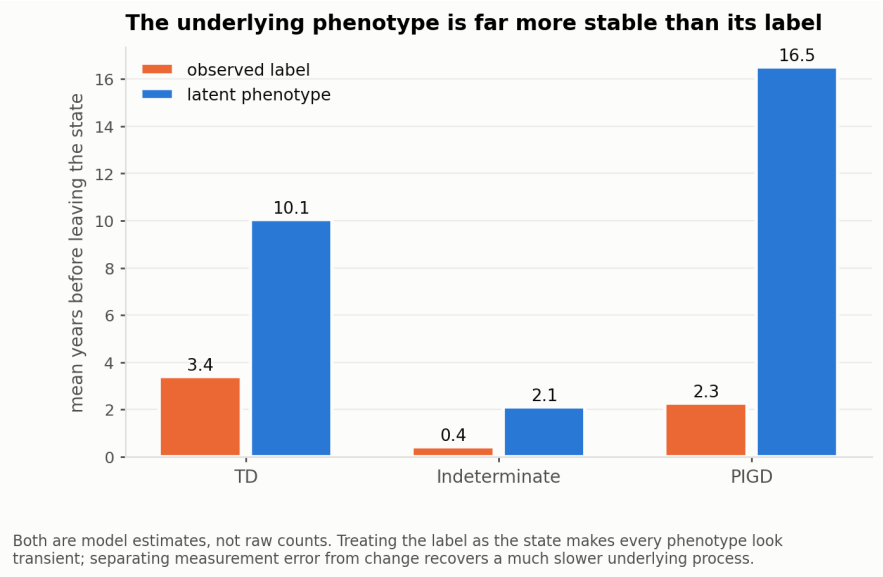


Figure 2: The same comparison. Both quantities are model estimates, not raw counts.

7 The deliverable: probabilities of subtype transition

Current	Horizon	P(TD)	P(Indeterminate)	P(PIGD)
TD	now	1.000	0.000	0.000
	1 year	0.910	0.071	0.019
	2 years	0.834	0.111	0.055
	3 years	0.769	0.133	0.099
	5 years	0.660	0.149	0.191
Indeterminate	now	0.000	1.000	0.000
	1 year	0.088	0.636	0.276
	2 years	0.137	0.424	0.440
	3 years	0.163	0.300	0.538
	5 years	0.182	0.185	**0.633**
PIGD	now	0.000	0.000	1.000
	1 year	0.003	0.047	0.950
	2 years	0.009	0.075	0.916
	3 years	0.018	0.091	0.891
	5 years	0.036	0.107	0.857

Table 10: Probability of occupying each subtype at each horizon, given the current phenotype. This is the output format specified in the proposal.

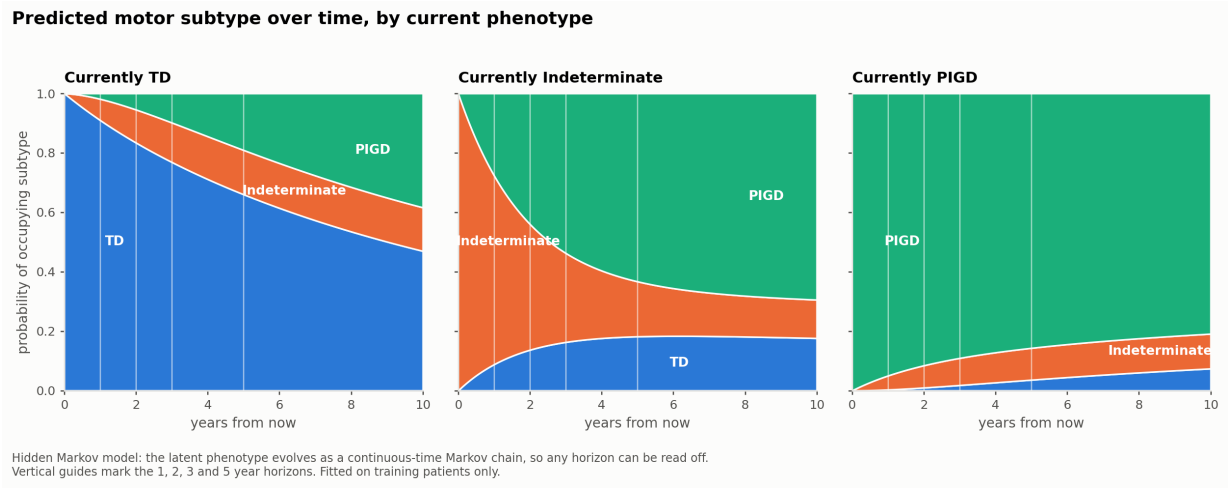


Figure 3: The same information continuously. Because the latent process runs in continuous time, any horizon can be read off, not only the four tabulated.

The clinically useful line. A patient currently Indeterminate has a 63% chance of being PIGD within five years, against 19% for a patient currently TD. Indeterminate is not a mild or intermediate finding — on these estimates it is the strongest short-term predictor of conversion in the model.

7.1 Calibration

Predicted-probability bin	n	Mean predicted	Observed rate
0.048–0.067	204	0.063	0.069
0.067–0.083	207	0.075	0.101
0.083–0.225	190	0.117	0.121
0.225–0.643	202	0.445	0.391
0.643–0.810	199	0.755	0.789

Table 11: Held-out calibration for P(PIGD at next visit). Predictions track observed frequencies across the range; the model is not systematically over- or under-confident.

8 Limitations

- **The emission floor is an assumption, not an estimate.** The Indeterminate row rests on it. The honest statement is that at most half of Indeterminate labels are correct; the data alone would say one third.
- **Time-homogeneity is assumed and is violated.** Fitting the continuous-time chain separately to early and late follow-up gives a TD → PIGD rate of 0.045 per year before year 3 and 0.085 after — nearly double. The model averages over disease duration.
- **No covariates.** The model uses only the subtype sequence. None of the baseline predictors assembled in Weeks 1–2 (RBDSQ, SCOPA-AUT, UPSIT, MoCA, DaTscan, age) enter it yet.
- **Three states, not a continuous ratio.** This is a discrete approximation to the proposal's design. 16.3% of labelled visits have no finite ratio, which is one reason the discrete form was built first.
- **Label availability is not random.** Carried over from Weeks 3–4: visits are lost preferentially for patients who cannot complete the pull test, who are more advanced. Progression estimates here remain lower bounds.
- **Held-out, not prospective.** Validation is a random split of one cohort. Nothing here has been tested on an external cohort.

9 Anticipated questions

Q. The proposal asked for a Markov model. Why did you build three?
Because the first two failed validation, and the failure was diagnosable. The discrete chain is the model as sketched; the continuous-time chain was the expected fix given the Weeks 3–4 interval finding; neither beat the other meaningfully because both are misspecified in the same way. The Markov test in Section 3 identified the reason, and the HMM is the model that reason points to.

Q. Isn't a 14-parameter model beating a 6-parameter one just overfitting?
The comparison is on *held-out patients* who were not used in fitting, and it is repeated on 10 independent splits with a win in every one. Overfitting would show as an advantage on training data that reverses out of sample. It does not reverse.

Q. Why did the continuous-time model lose? Weeks 3–4 said the interval matters.
The interval does matter — that finding stands. But $P(t) = \exp(Qt)$ imposes a strong consistency condition across intervals, and a process observed with 35% immediate reversion does not satisfy it. Imposing a correct-looking constraint on a misspecified model makes it worse, which is why the loss is largest at long intervals.

Q. How do you know the latent states mean anything?
Two independent checks. First, the model recovers an ordering it was never given: direct TD → PIGD transitions are 16 times rarer than transitions via Indeterminate, which is what a continuous underlying ratio implies. Second, the estimated label reliability ranks Indeterminate lowest, matching the Weeks 3–4 reversion analysis computed by an entirely different method.

Q. Can this predict when a patient will convert?
Partly. The table gives the probability of being PIGD at any horizon for a patient in a given state, which answers "how likely, by when". It cannot yet give a patient-specific answer, because no patient-level covariates or random effects are in the model. That is the next stage.

Q. Is the model Markov in the latent state? Have you tested that?
No, that has not been tested. The Markov property was tested and rejected for the *observed* label. The HMM assumes the *latent* process is Markov, and its better held-out performance is consistent with that but does not establish it. A latent-state Markov test is a reasonable next check.

10 What comes next

Patient-specific progression
The current model gives one set of rates for the whole cohort. The proposal's hierarchical structure — a population trajectory plus subject-specific random effects — is what turns this into an individualised prediction.

Move to the continuous ratio
The three-state model is a discrete approximation. The proposal's design treats the observed ratio as a noisy reading of a latent continuous phenotype, which subsumes this model and resolves the Indeterminate identifiability problem, since a threshold band is not a state.

Relax time-homogeneity
Rates are demonstrably not constant over disease duration; allowing them to depend on time since diagnosis is a contained extension.

Add baseline covariates
The malignant-phenotype markers are already assembled and 97–100% complete at baseline. This is the second half of the proposal: does baseline diffuse-synucleinopathy burden predict progression, and progression *stability*?

Reproducing this report.

```
python scripts/03_progression_model.py     # ~20 min: 10 splits + bootstrap
python -m pytest tests/ -q                 # 118 tests
```

Stage 3 writes 14 CSV tables, 5 figures and `report.txt` to `outputs/model/`. Every number quoted here appears there.